

Real analysis Qualifying exam, August 2019

Problem 1

Let $\mathcal{C}$ be the Cantor set on $[0, 1]$. Recall that it is obtained by iteratively deleting the open middle third: $(\frac{1}{3}, \frac{2}{3})$, then $(\frac{1}{9}, \frac{2}{9}) \cup (\frac{7}{9}, \frac{8}{9})$, and so on.

(a) Show that $\mathcal{C} + \mathcal{C} := \{a + b : a, b \in \mathcal{C}\}$ is the full segment $[0, 2]$.

(b) Find two sets $A, B \subset \mathbb{R}$, each of which is closed and has Lebesgue measure zero, such that $A + B = \{a + b : a \in A, b \in B\}$ is the full line $\mathbb{R}$.

Problem 2

Does there exist a measure space $(X, \mathcal{F}, \mu)$ with a finite measure μ , and a sequence of μ -measurable functions $\{f_n\}_{n=1,2,\dots}$ on X such that:

- $f_n(x) \geq 0$ for all n, x ;
- $f_n(x) \rightarrow 0$ as $n \rightarrow +\infty$ for all x ;
- $\int f_n(x) \mu(dx) \rightarrow 0$ as $n \rightarrow +\infty$;
- $\Phi(x) := \sup_n f_n(x)$ has infinite integral?

If yes, give an example of such a sequence $\{f_n\}$. If no, give a proof of nonexistence.

Problem 3

Let μ be a signed Borel measure on $\mathbb{R}^n$ which is bounded on bounded sets. Suppose that $\int f d\mu = 0$ for all continuous functions f with bounded support. Show that then $\mu = 0$.

Problem 4

Let $L^1(\mathbb{R})$ be the space of Lebesgue integrable functions on $\mathbb{R}$. For a positive function $f \in L^1(\mathbb{R})$ show that the function $\frac{1}{f(x)}$ does not belong to $L^1(\mathbb{R})$.

(Hint: look at the function $1 = f^{1/2} f^{-1/2}$.)

Problem 5

Applying the Gram-Schmidt orthogonalization to $1, x, x^2, \dots$ in the Hilbert space $L^2([-1, 1])$ (with Lebesgue measure), one gets the Legendre polynomials $L_n(x), n = 0, 1, 2, \dots$

(a) Show that the Legendre polynomials form a basis (= complete orthogonal system) in the Hilbert space $L^2([-1, 1])$

(b) Show that the Legendre polynomials are given by the formula $L_n(x) = c_n \frac{d^n}{dx^n} (x^2 - 1)^n$ (you do not need to specify c_n).

(Hint: employ integration by parts.)